

Our Periodic Table

The first 20 elements, which make up almost everything around you.
The **bold** ones are in your molecule kit. Each row down adds one more shell of electrons.

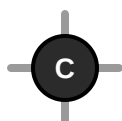
1 1 H Hydrogen 1.008 1 hole Build: H₂, H₂O, CH₄	8 2·6 O Oxygen 16.00 The air we breathe	How to read a square (Oxygen): 8 = atomic number: protons in the middle (and electrons around it) 2·6 = electrons in each shell, inside to outside O = symbol 16.00 = atomic weight = its color in your model kit Bold squares are in your kit. Faded ones are not (yet!)						2 2 He Helium 4.003 Party balloons
3 2·1 Li Lithium 6.94 Batteries	4 2·2 Be Beryllium 9.012 Space telescope mirrors	5 2·3 B Boron 10.81 Slime (borax)	6 2·4 C Carbon 12.01 4 holes Build: CH₄, CO₂, benzene	7 2·5 N Nitrogen 14.01 3 holes Build: NH₃, glycine	8 2·6 O Oxygen 16.00 2 holes Build: O₂, H₂O, CO₂	9 2·7 F Fluorine 19.00 Toothpaste	10 2·8 Ne Neon 20.18 Glowing signs	
11 2·8·1 Na Sodium 22.99 Table salt	12 2·8·2 Mg Magnesium 24.31 Firework sparkles	13 2·8·3 Al Aluminum 26.98 Soda cans, foil	14 2·8·4 Si Silicon 28.09 Sand & computer chips	15 2·8·5 P Phosphorus 30.97 Match heads	16 2·8·6 S Sulfur 32.06 Rotten-egg smell	17 2·8·7 Cl Chlorine 35.45 1 hole Build: HCl	18 2·8·8 Ar Argon 39.95 Inside light bulbs	
19 2·8·8·1 K Potassium 39.10 Bananas	20 2·8·8·2 Ca Calcium 40.08 Bones & milk	Spot the pattern! Go left to right across the bold squares: C has 4 holes, N has 3, O has 2, Cl has 1. The closer an atom is to the right side, the fewer electrons it still needs to fill its outer shell. The noble gases at the far right are already full, so they don't bond at all. That's why they're loners!						

Your kit's atom team

Holes = bonds it can make



Hydrogen
1 hole
The connector.
Caps off the ends.



Carbon
4 holes
The builder. Makes chains & rings.



Nitrogen
3 holes
In the air & in your muscles.



Oxygen
2 holes
In water, air & CO₂.

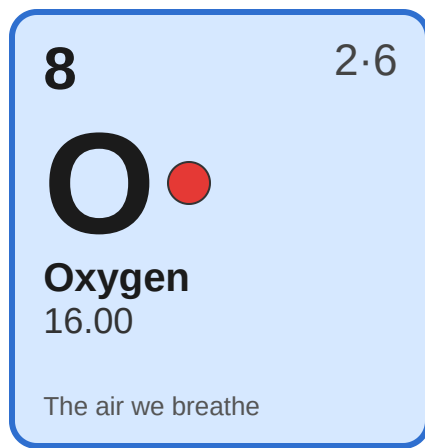


Chlorine
1 hole
Grabs 1 partner, like H.

The real table has **118** elements. Your kit may also have yellow sulfur (S) or purple phosphorus (P). Check the color card!

Zoom Into Water

From a drop, to a molecule, to the atoms inside it.



8

Atomic number

The number of **protons** in the middle of the atom. It is the atom's ID number: every oxygen atom has 8. A plain atom has the same number of **electrons** too.

2·6

Electron shells

Electrons zoom around in layers called shells. Oxygen has **2** in the first shell and **6** in the outer shell.

O

Symbol

A short nickname scientists everywhere use. Some come from Latin: gold is **Au** from *aurum*.

Oxygen

Name

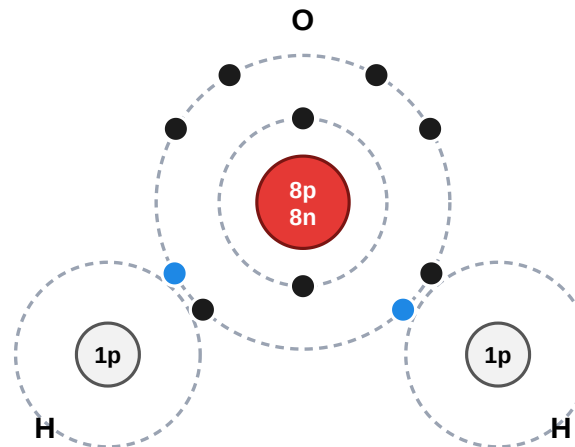
What we call it. Oxygen means "acid maker" in Greek.

16.00

Atomic weight

How heavy one atom is. Oxygen weighs about as much as **16 hydrogen atoms**. Weight minus atomic number = **neutrons**: $16 - 8 = 8$.

1 water molecule = 2 H + 1 O



● protons (p) & neutrons (n) in the middle ● oxygen's electrons ● electrons hydrogen shares
Oxygen wants **8** electrons in its outer shell but only has 6. Each hydrogen **shares** 1 electron, so now oxygen has 8. That sharing is the **bond**: the stick in your model kit!

How much does water weigh?

$$\text{H } 1 + \text{H } 1 + \text{O } 16 = 18$$

A water molecule weighs about as much as 18 hydrogen atoms.

Why do kit atoms have different numbers of holes?

Atom	Outer shell	Needs	Holes
○ H	1 of 2	1	1
● O	6 of 8	2	2
● N	5 of 8	3	3
● C	4 of 8	4	4

Each hole is an electron the atom still needs!